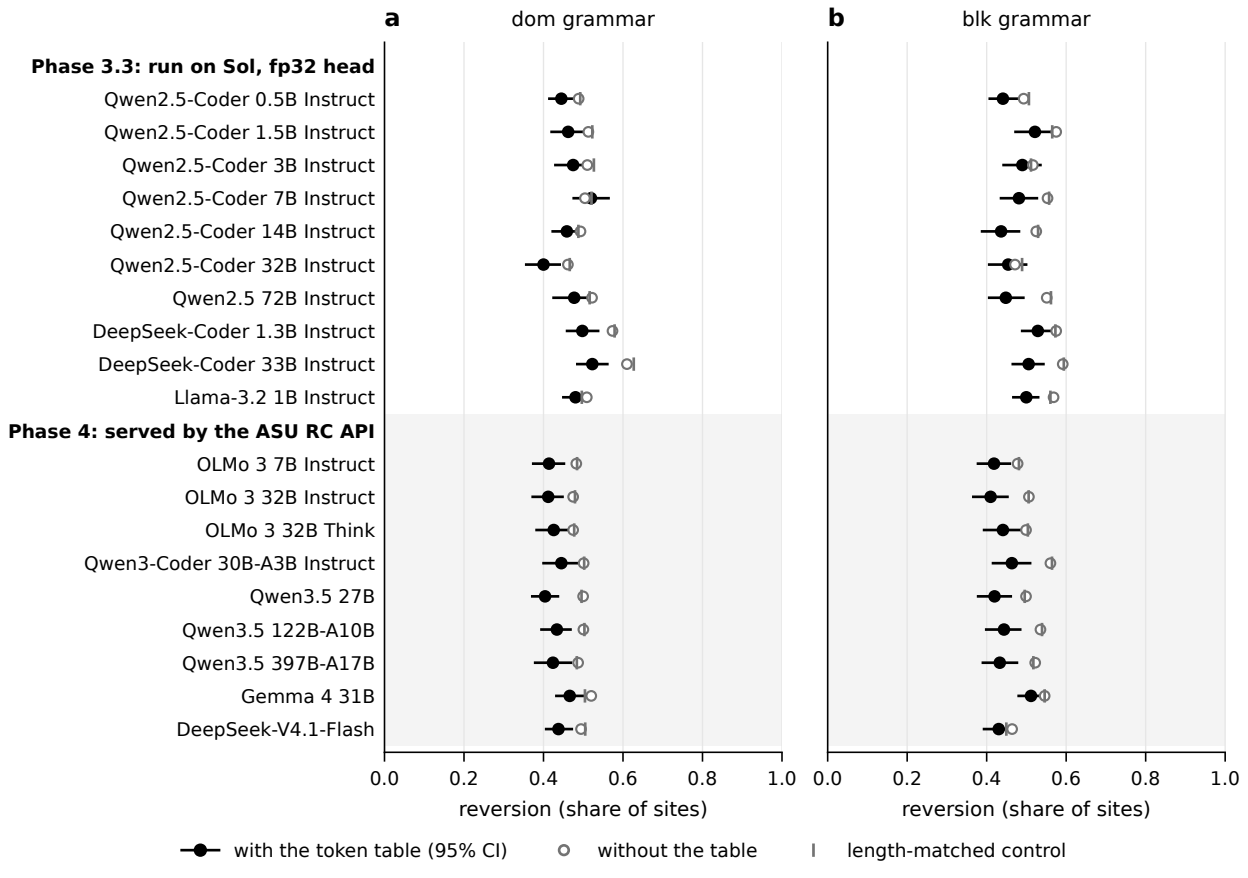
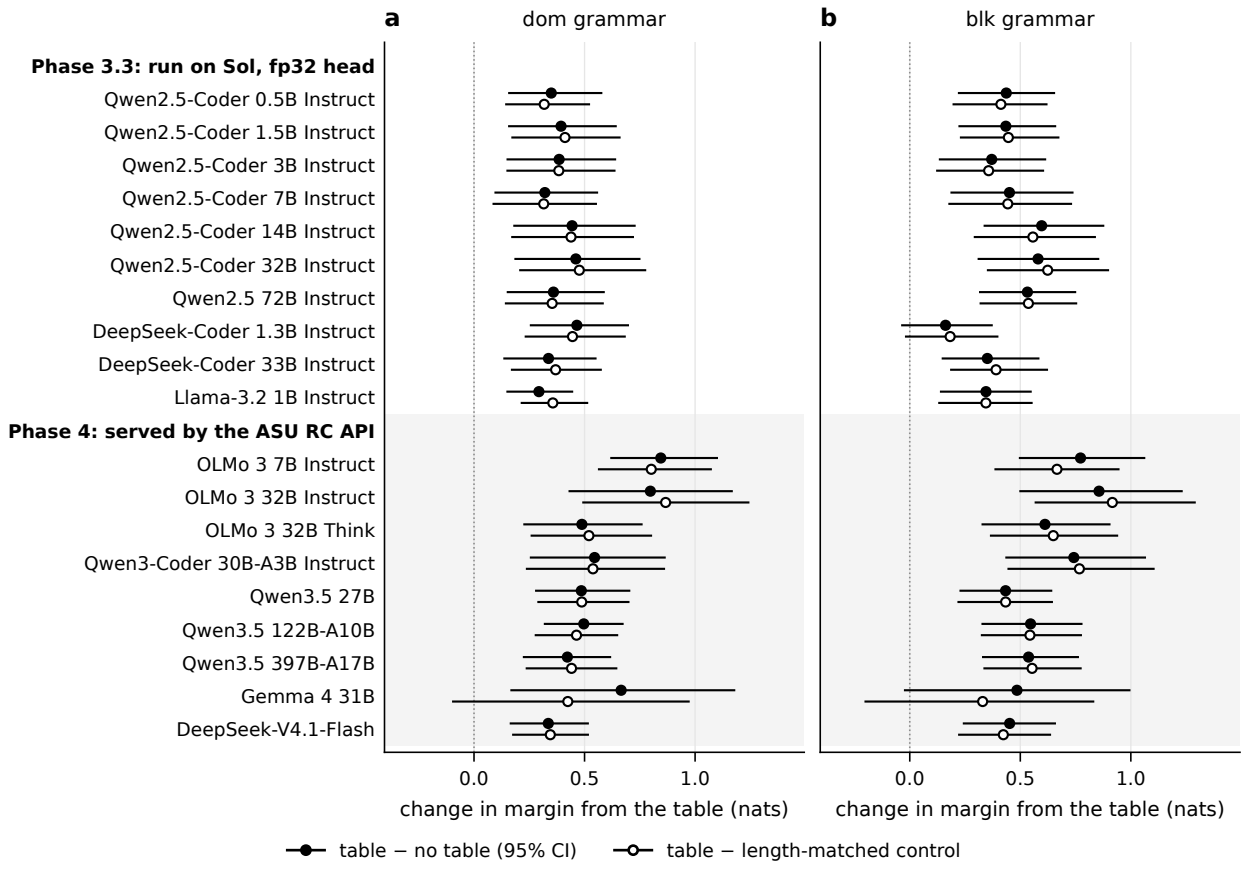
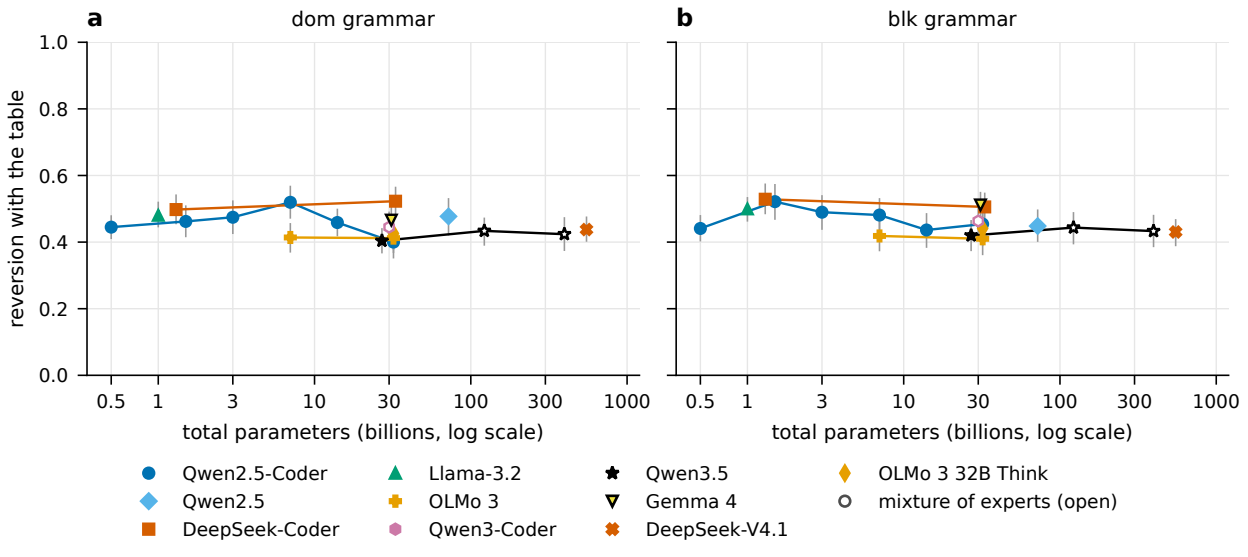
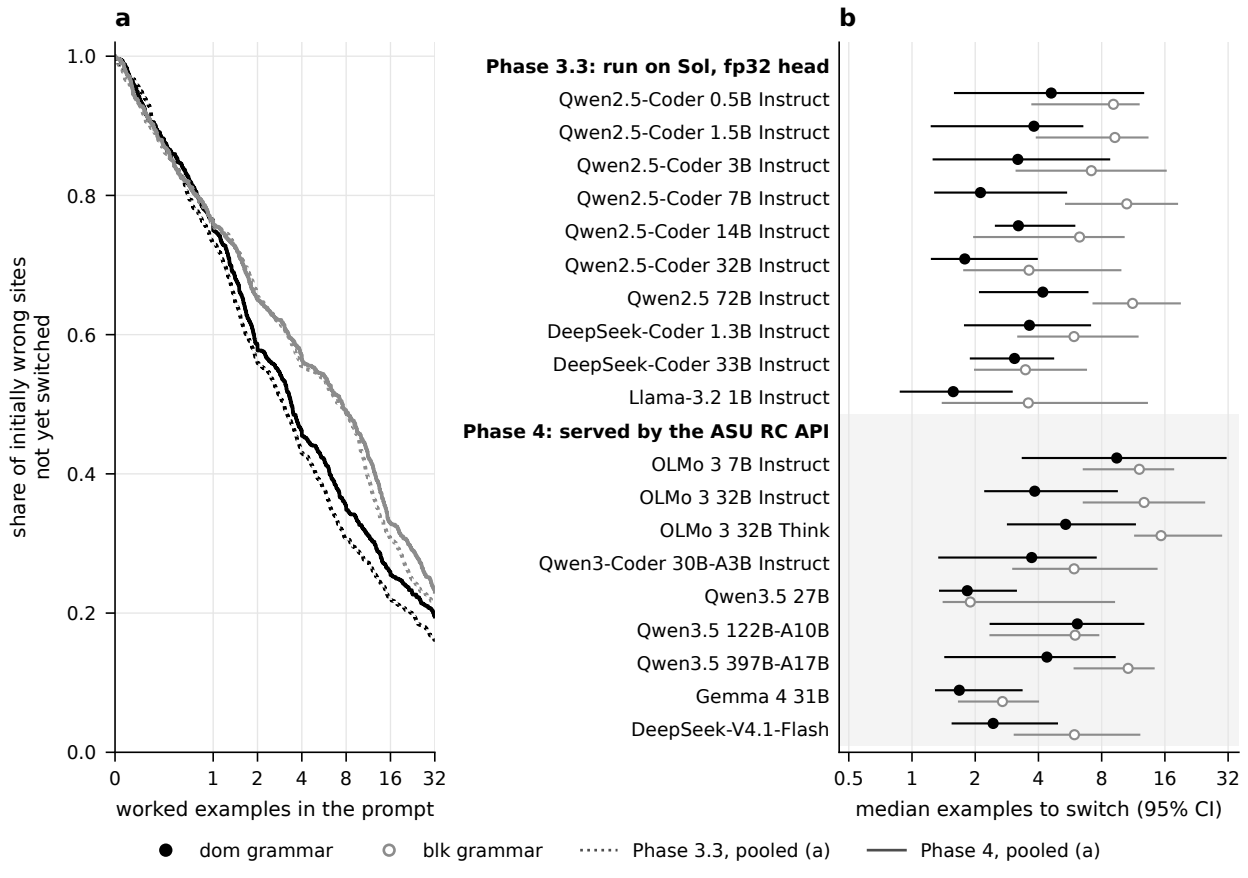
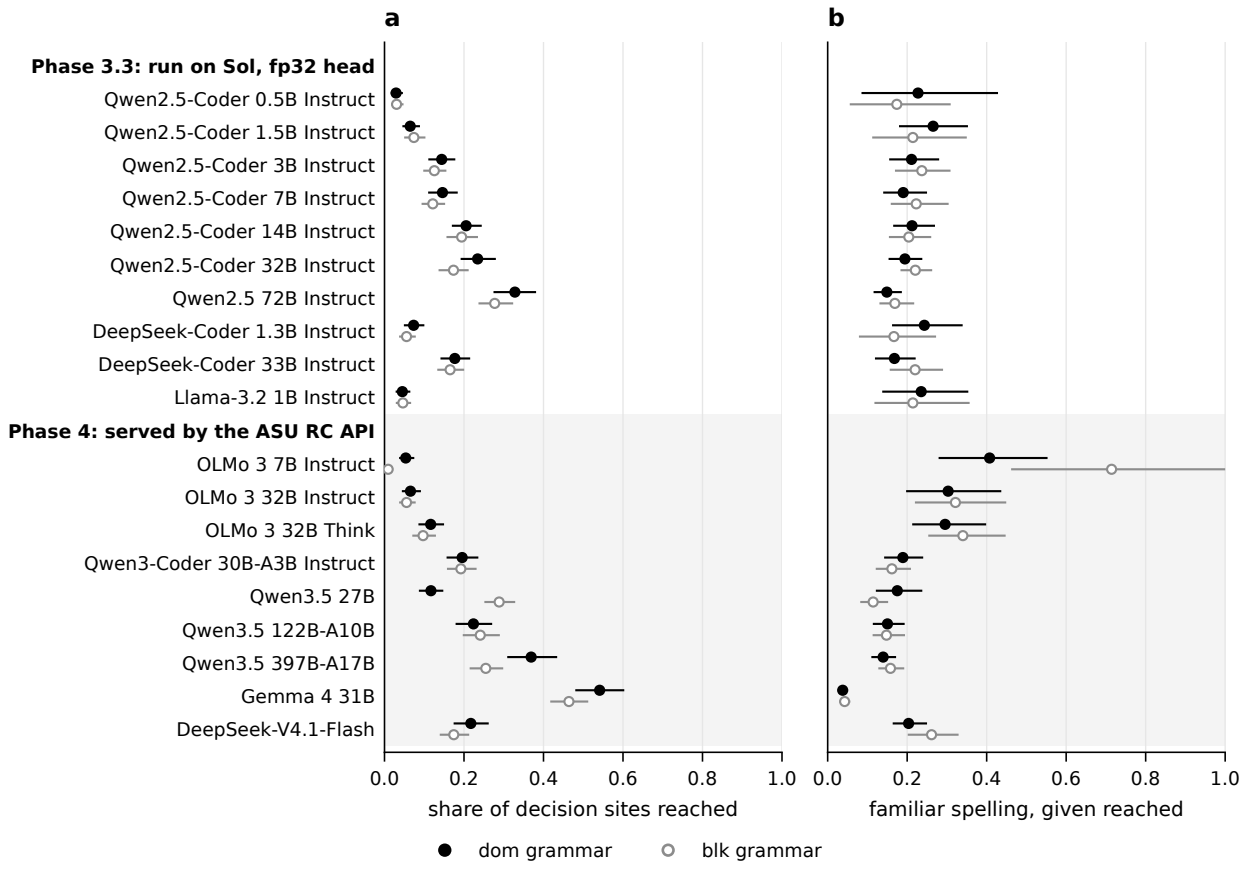
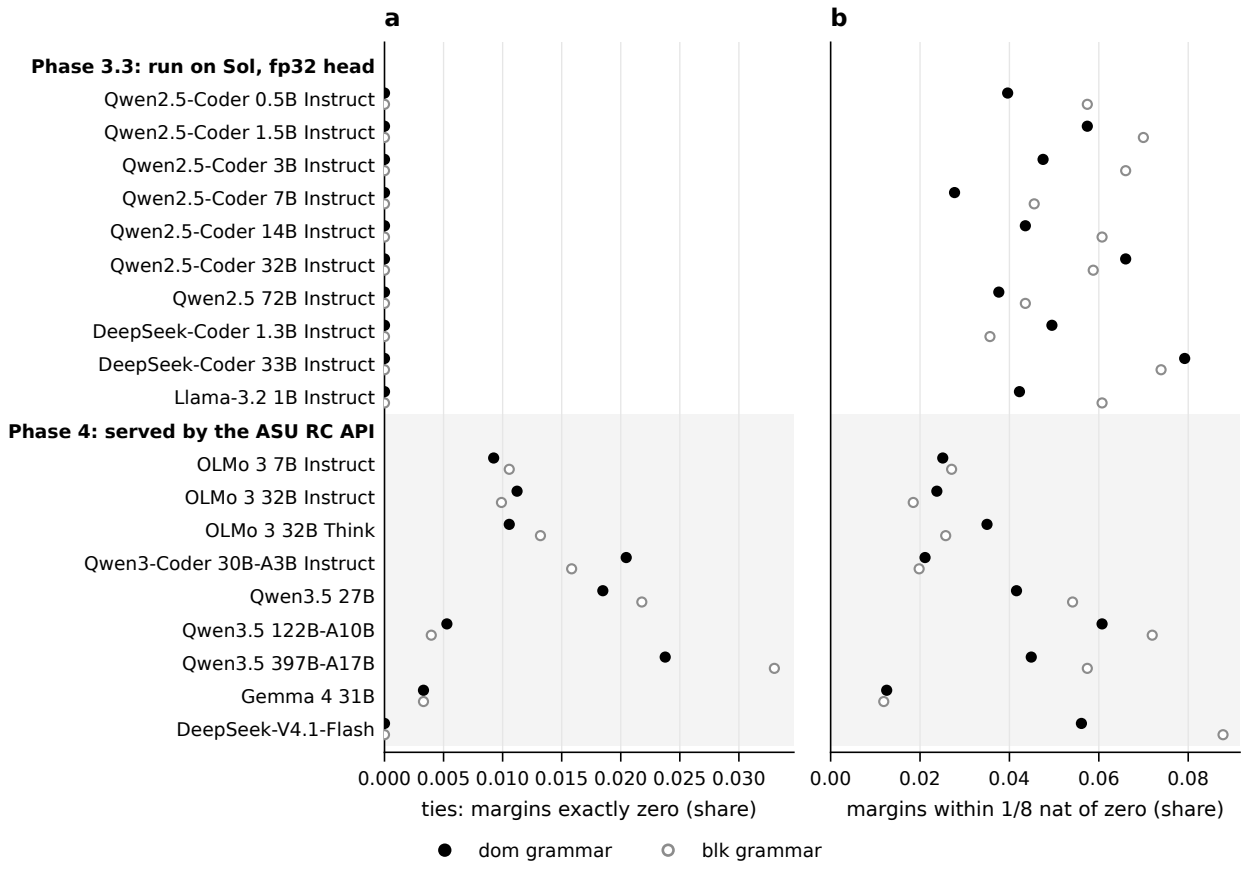

a
dom grammar
b
blk grammar
Phase 3.3: run on Sol, fp32 head
Qwen2.5-Coder 0.5B Instruct
Qwen2.5-Coder 1.5B Instruct
Qwen2.5-Coder 3B Instruct
Qwen2.5-Coder 7B Instruct
Qwen2.5-Coder 14B Instruct
Qwen2.5-Coder 32B Instruct
Qwen2.5 72B Instruct
DeepSeek-Coder 1.3B Instruct
DeepSeek-Coder 33B Instruct
Llama-3.2 1B Instruct
Phase 4: served by the ASU RC API
OLMo 3 7B Instruct
OLMo 3 32B Instruct
OLMo 3 32B Think
Qwen3-Coder 30B-A3B Instruct
Qwen3.5 27B
Qwen3.5 122B-A10B
Qwen3.5 397B-A17B
Gemma 4 31B
DeepSeek-V4.1-Flash
0.0
0.2
0.4
0.6
0.8
1.0
reversion (share of sites)
with the token table (95% CI)
without the table
length-matched control

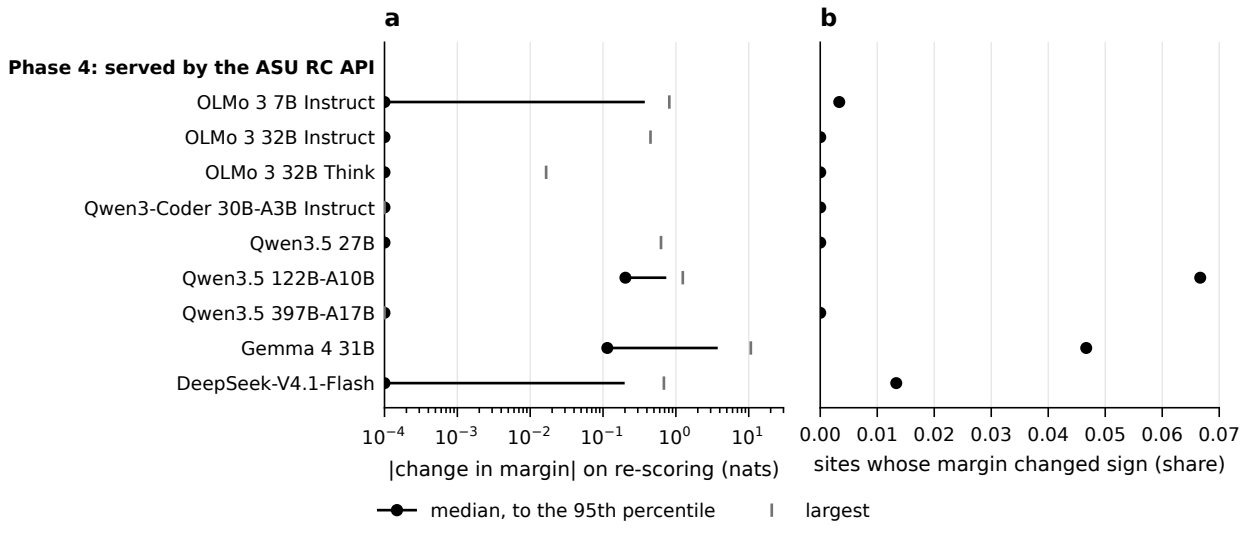

a
dom grammar
b
blk grammar
Phase 3.3: run on Sol, fp32 head
Qwen2.5-Coder 0.5B Instruct
Qwen2.5-Coder 1.5B Instruct
Qwen2.5-Coder 3B Instruct
Qwen2.5-Coder 7B Instruct
Qwen2.5-Coder 14B Instruct
Qwen2.5-Coder 32B Instruct
Qwen2.5 72B Instruct
DeepSeek-Coder 1.3B Instruct
DeepSeek-Coder 33B Instruct
Llama-3.2 1B Instruct
Phase 4: served by the ASU RC API
OLMo 3 7B Instruct
OLMo 3 32B Instruct
OLMo 3 32B Think
Qwen3-Coder 30B-A3B Instruct
Qwen3.5 27B
Qwen3.5 122B-A10B
Qwen3.5 397B-A17B
Gemma 4 31B
DeepSeek-V4.1-Flash
0.0
0.5
1.0
change in margin from the table (nats)
0.0
0.5
1.0
change in margin from the table (nats)
table − no table (95% CI)
table − length-matched control


a
dom grammar
b
blk grammar
reversion with the table
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1
3
10
30
100
300
1000
total parameters (billions, log scale)
Qwen2.5-Coder
Qwen2.5
DeepSeek-Coder
Llama-3.2
OLMo 3
Qwen3-Coder
Qwen3.5
Gemma 4
DeepSeek-V4.1
OLMo 3 32B Think
mixture of experts (open)


a
b
share of initially wrong sites not yet switched
1.0
0.8
0.6
0.4
0.2
0.0
0
1
2
4
8
16
32
worked examples in the prompt
Phase 3.3: run on Sol, fp32 head
Qwen2.5-Coder 0.5B Instruct
Qwen2.5-Coder 1.5B Instruct
Qwen2.5-Coder 3B Instruct
Qwen2.5-Coder 7B Instruct
Qwen2.5-Coder 14B Instruct
Qwen2.5-Coder 32B Instruct
Qwen2.5 72B Instruct
DeepSeek-Coder 1.3B Instruct
DeepSeek-Coder 33B Instruct
Llama-3.2 1B Instruct
Phase 4: served by the ASU RC API
OLMo 3 7B Instruct
OLMo 3 32B Instruct
OLMo 3 32B Think
Qwen3-Coder 30B-A3B Instruct
Qwen3.5 27B
Qwen3.5 122B-A10B
Qwen3.5 397B-A17B
Gemma 4 31B
DeepSeek-V4.1-Flash
0.5
1
2
4
8
16
32
median examples to switch (95% CI)
dom grammar
blk grammar
Phase 3.3, pooled (a)
Phase 4, pooled (a)


a
b
Phase 3.3: run on Sol, fp32 head
Qwen2.5-Coder 0.5B Instruct
Qwen2.5-Coder 1.5B Instruct
Qwen2.5-Coder 3B Instruct
Qwen2.5-Coder 7B Instruct
Qwen2.5-Coder 14B Instruct
Qwen2.5-Coder 32B Instruct
Qwen2.5 72B Instruct
DeepSeek-Coder 1.3B Instruct
DeepSeek-Coder 33B Instruct
Llama-3.2 1B Instruct
Phase 4: served by the ASU RC API
OLMo 3 7B Instruct
OLMo 3 32B Instruct
OLMo 3 32B Think
Qwen3-Coder 30B-A3B Instruct
Qwen3.5 27B
Qwen3.5 122B-A10B
Qwen3.5 397B-A17B
Gemma 4 31B
DeepSeek-V4.1-Flash
0.0
0.2
0.4
0.6
0.8
1.0
share of decision sites reached
familiar spelling, given reached
dom grammar
blk grammar


a
b
Phase 3.3: run on Sol, fp32 head
Qwen2.5-Coder 0.5B Instruct
Qwen2.5-Coder 1.5B Instruct
Qwen2.5-Coder 3B Instruct
Qwen2.5-Coder 7B Instruct
Qwen2.5-Coder 14B Instruct
Qwen2.5-Coder 32B Instruct
Qwen2.5 72B Instruct
DeepSeek-Coder 1.3B Instruct
DeepSeek-Coder 33B Instruct
Llama-3.2 1B Instruct
Phase 4: served by the ASU RC API
OLMo 3 7B Instruct
OLMo 3 32B Instruct
OLMo 3 32B Think
Qwen3-Coder 30B-A3B Instruct
Qwen3.5 27B
Qwen3.5 122B-A10B
Qwen3.5 397B-A17B
Gemma 4 31B
DeepSeek-V4.1-Flash
0.000 0.005 0.010 0.015 0.020 0.025 0.030
ties: margins exactly zero (share)
0.00 0.02 0.04 0.06 0.08
margins within 1/8 nat of zero (share)
dom grammar
blk grammar


a
b
Phase 4: served by the ASU RC API
OLMo 3 7B Instruct
OLMo 3 32B Instruct
OLMo 3 32B Think
Qwen3-Coder 30B-A3B Instruct
Qwen3.5 27B
Qwen3.5 122B-A10B
Qwen3.5 397B-A17B
Gemma 4 31B
DeepSeek-V4.1-Flash
10^{-4} 10^{-3} 10^{-2} 10^{-1} 10^{0} 10^{1}
|change in margin| on re-scoring (nats)
0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07
sites whose margin changed sign (share)
median, to the 95th percentile
largest